## 4.6.3 The development of understanding of genetics and evolution

### 4.6.3.1 Evidence for evolution

| Content | Key opportunities for skills development |
|---|---|
| Students should be able to describe the evidence for evolution including fossils and antibiotic resistance in bacteria.<br><br>The theory of evolution by natural selection is now widely accepted.<br><br>Evidence for Darwin's theory is now available as it has been shown that characteristics are passed on to offspring in genes. There is further evidence in the fossil record and the knowledge of how resistance to antibiotics evolves in bacteria. | WS 1.3<br><br>Data is now available to support the theory of evolution. |

### 4.6.3.2 Fossils

| Content | Key opportunities for skills development |
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| Fossils are the 'remains' of organisms from millions of years ago, which are found in rocks.<br>Fossils may be formed:<br>• from parts of organisms that have not decayed because one or more of the conditions needed for decay are absent<br>• when parts of the organism are replaced by minerals as they decay<br>• as preserved traces of organisms, such as footprints, burrows and rootlet traces. | MS 2c, 4a<br>Extract and interpret information from charts, graphs and tables. |
| Many early forms of life were soft-bodied, which means that they have left few traces behind. What traces there were have been mainly destroyed by geological activity. This is why scientists cannot be certain about how life began on Earth. | WS 1.3<br>Appreciate why the fossil record is incomplete. |
| We can learn from fossils how much or how little different organisms have changed as life developed on Earth. | WS 1.1<br>Understand how scientific methods and theories develop over time. |
| Students should be able to extract and interpret information from charts, graphs and tables such as evolutionary trees. | MS 2c, 4a |

### 4.6.3.3 Extinction

| Content | Key opportunities for skills development |
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| Extinctions occur when there are no remaining individuals of a species still alive.<br>Students should be able to describe factors which may contribute to the extinction of a species. | |

### 4.6.3.4 Resistant bacteria

| Content | Key opportunities for skills development |
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| Bacteria can evolve rapidly because they reproduce at a fast rate.<br><br>Mutations of bacterial pathogens produce new strains. Some strains might be resistant to antibiotics, and so are not killed. They survive and reproduce, so the population of the resistant strain rises. The resistant strain will then spread because people are not immune to it and there is no effective treatment. | |
| MRSA is resistant to antibiotics.<br><br>To reduce the rate of development of antibiotic resistant strains:<br>• doctors should not prescribe antibiotics inappropriately, such as treating non-serious or viral infections<br>• patients should complete their course of antibiotics so all bacteria are killed and none survive to mutate and form resistant strains<br>• the agricultural use of antibiotics should be restricted.<br><br>The development of new antibiotics is costly and slow. It is unlikely to keep up with the emergence of new resistant strains. | There are links with this content to Antibiotics and painkillers (page 37). |